Agnes & Co.

RETAIL ANALYTICS

PREPARED FOR

ROSSMANN

ENGAGEMENT

Daily store-sales forecasting

DELIVERABLE

Executive dashboard

MODEL

All Business Interactions, OLS on LogSales

BASIS

100 stores · 42,400 store-days

ROSSMANN · 100 STORES · JANUARY 2014 TO FEBRUARY 2015

Improving daily store-sales forecasts to support promotion, inventory and staffing decisions

Sales for any store on any day can be forecast before that day begins, using only information already known: the calendar, the store itself, and the promotion schedule. This dashboard sets out what the forecast reveals, where promotions pay and where they do not, what the analysis could not establish, and what management should do differently. Every figure is drawn from *Rossmann Store Sales Forecast* with its source section marked.

OBJECTIVE

Evaluate whether information known before the sales day can improve daily store-sales forecasts and support promotion, inventory and staffing decisions

DEPENDENT VARIABLE

LogSales, the natural log of daily turnover. Predictions are converted back to the Sales scale for operational interpretation

MODEL

All Business Interactions: OLS with store fixed effects, calendar and holiday controls, Promo2 and competition categories, and regular-promotion interactions. HC3 robust standard errors

SAMPLE

100 stores, 42,400 store-days, January 2014 to February 2015. 31,249 training and 5,131 test observations after retaining open stores with positive sales

VALIDATION

2014 for development, January–February 2015 as unseen chronological holdout. Training R^2 0.8323, test R^2 0.8293 on LogSales

TEST ERROR

RMSE 0.1851 and MAE 0.1331 on LogSales. 95% prediction intervals covered 95.30% of actual test observations

SUMMARY

What the analysis found

This project evaluated whether Rossmann could use information known before the sales day to improve daily store-sales forecasts and support promotion, inventory and staffing decisions. Every figure is drawn from the technical report with its source section marked; plain-English definitions are collected under Terms at the end.

01 Which store it is matters more than anything else we measured

EVIDENCE — Giving every store its own baseline lifted forecast accuracy from 28% to 81% — a bigger gain than every other improvement in the project combined. §3.4, §3.5.4

IMPLICATION — Forecast every store separately rather than applying one network figure. The store effect is not itself something management can pull — it stands in for location quality, local traffic, store size, demographics, customer loyalty, availability and how well the store is run.

02 Day promotions work, but nowhere near equally

EVIDENCE — A single-day promotion lifts sales by **78.7%** on a Monday but only **37.4%** by Friday. Lift also differs by store format, product range, school holidays and which programme plan a store is on. §3.5.4

IMPLICATION — Stop planning promotions on one chain-wide lift assumption. Match promotional stock and staffing to the specific store and day.

03

Two findings that are easy to read the wrong way

EVIDENCE

Sales look **2.8% lower** when the continuing promotion programme is running — but that comparison ignores which stores, which months and what else was on. Once those are accounted for, the programme does carry real forecasting value. Separately, distance to the nearest competitor turns out to predict nothing at all.

§3.3.3, §3.5.4

IMPLICATION

Keep the continuing programme, but judge it plan by plan rather than as a whole. And do not conclude competition does not matter — only that distance on its own tells us nothing useful.

04

The forecast is good enough to run inventory and rosters on

EVIDENCE

The model explains **83.2%** of daily sales variation in the period it was built on and **82.9%** on two months it had never seen — so it holds up on new data. Its planning ranges captured **95.3%** of what actually happened. §3.6.1, §3.6.2

IMPLICATION

Plan to the forecast as the central number and the range around it as the realistic spread, for both stock and staffing.

BEFORE GOING FURTHER

The two kinds of promotion

Rossmann runs two distinct promotional mechanics. They are frequently conflated in reporting, they behave differently in the analysis, and several of the findings below turn on the distinction.

§2.2, §3.2

MECHANIC ONE

The day promotion

A promotion a store runs on one specific trading day. It starts and ends with that day, and the store either runs one or it does not. This is the lever the promotional calendar controls.

Recorded in the data as **Promo**.

MECHANIC TWO

The continuing programme

A separate recurring programme a store signs up to and then stays in. New rounds launch on a fixed quarterly schedule, and stores sit on one of three cycles — **Plan A** relaunches in January, April, July and October; **Plan B** in February, May, August and November; **Plan C** in March, June, September and December.

Recorded in the data as **Promo2**. Enrolment is a standing commitment, not a calendar decision.

Why the difference matters. A store can run a day promotion whether or not it is enrolled in the continuing programme. The interesting question is what happens when both are live at once — and the analysis finds that the answer depends on which plan the store is on.

01

What matters most to daily sales

The forecast was built up in three steps, so we can see exactly where the accuracy came from. Almost all of it came from one change. §3.4

FORECAST ACCURACY ON UNSEEN DATA, BY BUILD STAGE

Business factors only	calendar, format, promotion	28% test R^2 0.2776
Plus store baselines	a separate level for every store	81% test R^2 0.8091
Plus promotion detail	how lift varies by day, format and range	83% test R^2 0.8293

> Knowing **which store** it is takes the forecast from 28% to 81% accurate. Everything we know about promotions, weekdays, formats, ranges, competition and the continuing programme together adds only **two further points.** a gain of 0.5315 in test R^2, §3.5.4

Worth stating plainly. The full model beats a simpler version — one using only promotion, weekday, format and range — by just **0.0012** of forecast accuracy. The extra detail on competition and the continuing programme earns its place for interpretation, not for prediction. §3.4

What this means in practice. The store effect is a placeholder for things we do not currently measure — catchment, footfall, floor space, local incomes, availability, and how well the store is run. It cannot be managed directly. Two things follow: forecast each store separately, and start collecting the data that would explain what the store effect actually stands for.

02

Where day promotions work hardest

Day promotions still drive sales once store, calendar and holiday differences are taken out. But how much they drive depends heavily on when they run and where — and the differences are large enough to be worth planning around. §3.3.2, §3.5.4

RAW COMPARISON

32.1% higher on promotion days

Stores average €8,993 on a promotion day against €6,806 otherwise. Before any adjustment, that gap already narrows from 50.0% on Monday to 21.0% on Friday.

LIKE FOR LIKE

78.7% Monday, 37.4% Friday

Comparing a store against itself on a comparable non-promotion day, the lift more than halves across the trading week.

WEEKEND

No weekend evidence

Day promotions were never run on a Saturday or Sunday in this period, so we cannot say what one would return.

SALES LIFT BY WEEKDAY

Table 6 of the technical report.

Monday	78.7%
Tuesday	61.7%
Wednesday	61.3%
Thursday	47.7%
Friday	37.4%

SALES LIFT BY STORE FORMAT AND RANGE

Formats A–D are StoreType; basic and extra are Assortment A and B. Table 7 of the technical report.

Format A	78.7%
Format B	60.7%
Format C	56.2%
Format D	61.2%
Basic range	78.7%
Extra range	42.5%

BEFORE ADJUSTMENT: RAW SALES LIFT BY CONDITION

What the unadjusted comparisons showed, before store, calendar and holiday differences were controlled. The modelled figures above are the like-for-like versions. §3.3.2

Format A	48.5%
Format B	18.2%
Basic range	41.0%
Extended range	31.5%
Extra range	7.3%
Competitor far	39.7%
No competitor	33.8%
Competitor near	26.5%

Why the two sets differ. Raw lift compares promotion days against non-promotion days without accounting for which stores ran them, in which months, or alongside what else. The modelled figures compare each

store against itself under matched conditions, which is why they are larger and why they are the ones to plan against.

What to do with this. Put more promotional stock and staffing behind Monday to Wednesday, and stop assuming one lift figure holds all week. Set promotion intensity by format and range rather than centrally, and look specifically at how promotions are designed and executed in extra-range stores, which respond least. One clarification worth stating: because each store already has its own baseline, these figures compare how differently stores *respond to promotion* — not how much they sell.

03

Calendar, competition and the continuing programme

Three more questions were put to the model. Two produced effects worth planning around. One produced nothing – which is itself worth knowing. §3.5.4

CALENDAR — USE IT

December runs 28.7% above January

Seasonality is strong and reliable. Public holidays run about **24.6%** above a normal day for stores that open. Easter is up 14.7% but less certain, and Christmas shows a 11.3% dip the model cannot confirm. School holidays register a real but small effect (p = 0.0039).

So: keep seasonality in the forecast, build extra stock and staffing into December, and remember holiday figures reflect only the stores permitted to open.

COMPETITION — NOT USABLE

How close a rival is tells us nothing

Once each store has its own baseline, competitor distance adds nothing to the forecast (p = 0.400). The raw numbers are revealing too: stores with no rival at all average €9,075, but those with one *nearby* average €8,302 against €7,094 for those with one further away — the opposite of what you would expect, and probably because busy locations attract competitors. Near and far are split at **1,630 metres**, the sample median.

So: stop using proximity as an explanation for performance. Competition may well matter — distance is simply the wrong measure of it.

CONTINUING PROGRAMME — KEEP IT

Valuable, despite how the raw numbers look

On the face of it, sales are **2.8% lower** while the programme is running. But that compares different stores in different months with different offers on. Once those are held constant, the programme does add forecasting value — and it changes how well day promotions perform. Stores on **Plans A and C** get more from a day promotion; stores on **Plan B** get less.

So: keep the programme in the forecast, but stop judging it as one thing. Measure each plan separately.

04

Planning inventory and staffing

Promotion is the most visible use of the forecast, but not the only one. Because every input is known before the trading day, the same forecast can be used to size stock and shifts weeks ahead. §3.2, §3.6

THE CENTRAL NUMBER

Plan to the forecast

Use the point forecast as the expected sales figure for a given store on a given day, and size replenishment and shifts against it. On ordinary trading days it is stable enough to plan on directly.

THE SPREAD

Size the buffer to the range

Around every forecast sits a planning range. Its lower and upper limits are what stock cover and shift cover should be set against — not the central number alone. Treat them as decision-support bounds, not guaranteed minimums or maximums.

THE EXCEPTIONS

Override on unusual days

Very high and very low days remain the hardest to predict. Where a store team already expects something out of the ordinary, their knowledge should take precedence over the forecast.

WHERE THE FORECAST CHANGES WHAT YOU WOULD OTHERWISE PLAN

Four patterns the model identifies well in advance, each with a direct operational consequence.

KNOWN IN ADVANCE	EFFECT ON SALES	WHAT IT MEANS FOR STOCK AND SHIFTS
December	+28.7%	Largest seasonal build of the year. Plan replenishment and additional hours well ahead of the month, not into it.
Public holidays	+24.6%	Strong days for the stores permitted to open. Check which stores those are before reading the figure across the estate.

KNOWN IN ADVANCE	EFFECT ON SALES	WHAT IT MEANS FOR STOCK AND SHIFTS
Promotion, Monday to Wednesday	+78.7% to +61.3%	The heaviest demand days in the week. Weight promotional stock and cover here.
Promotion, Thursday to Friday	+47.7% to +37.4%	Materially lighter. Holding the same cover as Monday over-resources the back half of the week.

The report's recommendation. Allocate more promotional inventory and staffing support to Monday through Wednesday, and do not assume the same uplift applies throughout the week. Because no regular promotions were observed on Saturday or Sunday, no weekend effect can be estimated. §3.5.4

05

Using the forecast to plan

The technical report establishes that point forecasts can serve as central daily estimates for inventory, staffing and promotional planning, and that prediction intervals provide realistic planning ranges. This tool applies the fitted model in that way. Every input is known before the trading day, so a month can be forecast in advance. Click any weekday to place or remove a regular promotion. §3.6

About the sales index. The model was fitted on 2014–15 trading and contains no growth trend, so it cannot know what has happened to sales volumes since. What it does know is *shape*: seasonality, weekday response and promotion lift are all multiplicative, so the ranking of promotion days — and therefore the optimal calendar — is unchanged by growth. Set the index to your current turnover against the 2014–15 baseline (100 means unchanged, 140 means trading 40% higher) and every euro figure rescales. Leave it at 100 and read the figures as relative rather than absolute.

PLANNING FOR: Entire sample — 100 stores

MONTH: September

YEAR: 2026

PROMOTION DAYS AVAILABLE: 11

SALES INDEX: 100

Rank by forecast | Clear

Stores 100 In Promo2 37 Trading days 26

MON	TUE	WED	THU	FRI	SAT	SUN
	1 €611k +€249k avail.	2 €569k +€230k avail.	3 €611k +€175k avail.	4 €670k +€132k avail.	5 NO PROMO €575k	6 NO PROMO €173k

7 €642k +€357k avail.	8 €611k +€249k avail.	9 €569k +€230k avail.	10 €611k +€175k avail.	11 €670k +€132k avail.	12 NO PROMO €575k	13 NO PROMO €173k
14 €642k +€357k avail.	15 €611k +€249k avail.	16 €569k +€230k avail.	17 €611k +€175k avail.	18 €670k +€132k avail.	19 NO PROMO €575k	20 NO PROMO €173k
21 €642k +€357k avail.	22 €611k +€249k avail.	23 €569k +€230k avail.	24 €611k +€175k avail.	25 €670k +€132k avail.	26 NO PROMO €575k	27 NO PROMO €173k
28 €642k +€357k avail.	29 €611k +€249k avail.	30 €569k +€230k avail.				

A Promo2 round is active this month for 8 of 100 stores. Plans A and C are associated with a stronger regular-promotion response and Plan B with a weaker one; the forecast accounts for this.

FORECAST TURNOVER, THIS PLAN

€16.58m

95% range €16.47m to €16.70m

INCREMENTAL FROM PROMOTIONS

€0

0.0% above an unpromoted month

PROMOTION DAYS USED

0 of 11

BEST POSSIBLE ARRANGEMENT

€3.14m

place some days to compare

> Place 11 promotion days on the calendar, or press **Rank by forecast** to see where the model would place them.

Weekday comparison

The same number of promotion days can be worth substantially different amounts depending on which weekdays they land on. This is your current plan against the best available arrangement, by weekday.

WEEKDAY	VALUE PER DAY	YOUR PLAN	BEST PLAN	DIFFERENCE
Monday	€357k	0	4	+4
Tuesday	€249k	0	5	+5
Wednesday	€230k	0	2	+2
Thursday	€175k	0	0	–
Friday	€132k	0	0	–

Forecasting a single day

The same model applied to one store on one date, with and without a regular promotion. Store format, product range, competition and programme plan are read from the store record. §3.6.1

STORE

Store 16 — StoreType A

Format, range and coupon cycle are read from the store record.

MONTH

September

WEEKDAY

Monday

THIS STORE

Store type **StoreType A** Assortment **Assortment C (extended)** Competition **Active far**

Promo2 **not participating**

INCREMENTAL TURNOVER, THIS ONE DAY

€3,990

Without a promotion	€6,516
With a promotion	€10,506
Response rate	61.2%
Rank across the estate	top 11%

Worth running. This day sits in the strongest quarter of everything available. Protect it in the calendar.

What the model cannot tell you

This forecast supports decisions; it does not make them. It is built for stores we already trade, under conditions like those we have seen. Six things it cannot do. §IV

COVERAGE

It covers 100 stores

These stores may not behave like the rest of the network. Treat the findings as strongly indicative rather than settled for every location.

NEW STORES

It cannot forecast a new store

The model learns each store's level from its own history. A store that has not traded yet has no history to learn from.

CAUSALITY

It shows patterns, not causes

Some of the promotion lift may reflect that we schedule promotions when we already expect a strong day. The figures should not be read as sales created purely by the promotion.

MISSING DATA

It knows nothing about margin

Pricing, discount depth, promotion cost, stock, stockouts, product mix, weather and local traffic were all unavailable. Every figure here is turnover, not profit.

EXTREMES

Exceptional days remain hard

Very high and very low sales days are less predictable than ordinary ones, and the widest planning ranges still missed a small number of them — the 99% range captured 97.74% rather than 99%. Use judgment on days you already expect to be unusual.

METHOD

It favours clarity over complexity

The method was chosen so every effect can be explained to a manager. That trade-off means it may miss tipping points and more tangled relationships. It also applies only to open trading days.

HOW THE FORECAST WAS CHECKED

Three tests were run on the model's errors to establish where it can be trusted and where it cannot. §3.5.1–3.5.3

TEST	WHAT IT FOUND	WHAT WAS DONE ABOUT IT
Spread of errors	Errors are not equally sized across the range – they narrow as	The statistical tests were re-run using a method that corrects for this. Forecasts and accuracy figures are unaffected; only the confidence in individual findings changes.

TEST	WHAT IT FOUND	WHAT WAS DONE ABOUT IT
	forecast sales rise.	
Shape of errors	Ordinary days sit close to the expected pattern, but both extremes depart from it, with one very large negative error.	With 31,249 training days the pattern is not a concern for the conclusions. It is the reason exceptional days carry wider ranges.
Unusual observations	1,507 store-days flagged for review against a very low threshold. The most extreme is Store 652 on 22 November 2014, which recorded sales of just 46.	None was removed. The largest influence score was 0.0175 against a conventional benchmark of 1, so no single day distorts the model. Flagged days should be checked against operational records.

Why this matters commercially. Nothing was quietly deleted to make the model look better. The 1,507 flagged days are a review list, not a cleanup list — several may be genuine closures, stock shortages or system faults worth investigating in their own right.

07

What to do differently

Eight changes the analysis supports, each with the finding that backs it. §3.5.4, §IV

01

Forecast each store separately

Replace network averages with a forecast for each store. What separates a strong location from a weak one is not its format or its competition — it is the location itself, and that only shows up store by store.

BECAUSE

Knowing the store takes forecast accuracy from **28%** to **81%**

02

Find out what the store effect is made of

The store effect is the single biggest thing in the model and we cannot see inside it. Start capturing footfall, catchment, floor space, capacity, availability and management data so the gap between our best and worst stores becomes something we can act on.

BECAUSE

Store differences carry more forecasting power than every business factor combined

03

Shift promotion support to earlier in the week

Move promotional stock and staffing toward Monday to Wednesday. The same promotion, run later in the week, returns roughly half as much.

BECAUSE

Uplift falls from **78.7%** on Monday to **37.4%** by Friday

04

Plan promotions by format and product mix

Set promotion intensity by store format and product range instead of centrally. Look hardest at extra-range stores, where promotions return least — the issue there may be the offer design rather than the discount.

BECAUSE

Lift ranges from **78.7%** in the strongest segment to **42.5%** in the weakest

05

Build the calendar into inventory and rosters

Keep seasonality in the forecast and build December into stock and rota planning well ahead. Where holiday figures look strong, check how many stores were actually open before reading across.

BECAUSE

December runs **28.7%** above January and public holidays **24.6%** above a normal day

06

Do not act on competitor distance

Proximity to a rival should stop being used to explain a store's performance or to justify where resource goes. If competition matters — and it may — we need data on rival size, pricing, quality and local footfall before we can say so.

BECAUSE

Competitor distance adds **nothing measurable** to the forecast

07

Keep the continuing programme, but judge it plan by plan

The programme improves our forecasts and changes how well day promotions perform, but it does not raise sales on its own. Measure each plan separately rather than reporting the programme as one number.

BECAUSE

Plans A and C strengthen day-promotion response; Plan B weakens it

08

Plan to a range, not a single number

Plan to the forecast as the expected number and the range around it as the realistic spread. Neither is a floor or a ceiling — use them alongside what the store team already knows about the week ahead.

BECAUSE

Planning ranges captured **95.3%** of what actually happened

> This is a strong forecasting baseline — but it is **a decision-support tool, not a demand-management system**. It tells us what demand is likely to be; it cannot yet tell us how to shape it. Adding pricing, discount depth, promotion cost, customer behaviour, stock availability and margin would close that loop: target an offer, observe the response, and update the next decision from what happened.

REFERENCE

Terms used on this page

Plain-English names used throughout, with the term each one corresponds to in the technical report.

OTHER TERMS USED ON THIS PAGE

Store baseline

STORE FIXED EFFECTS

The persistent trading level of a location once calendar and promotion effects are stripped out.

Store format

STORETYPE A–D

Rossmann's own store model categories. The data does not document what distinguishes them.

Product range

ASSORTMENT A, B, C

Basic range (A), extra range (B) and extended range (C).

Sales lift

PROMOTION UPLIFT

How much a day promotion raises sales above the same store on a comparable non-promotion day.

Planning range

PREDICTION INTERVAL

The range around a forecast within which actual sales are expected to fall.

Forecast accuracy

TEST R^2

The share of day-to-day variation in sales the model explains on data it has never seen.

RELATIONSHIP TO THE TECHNICAL REPORT

Every figure on this page is taken from *Rossmann Store Sales Forecast*, with its source section marked beside it. No variable has been added, no specification altered, and no analysis performed outside that document.

The forecasting tool in Section 04 evaluates the report's fitted All Business Interactions model. Its outputs are that model's own predictions applied to future dates.

THE MODEL

OLS on LogSales across 100 stores, January 2014 to February 2015, with store fixed effects, month, day of week, state and school holidays, Promo2 category and competition exposure, plus regular-promotion interactions with weekday, StoreType, Assortment, competition exposure, school-holiday status and Promo2 plan.

Estimated on 2014 data and tested on the unseen January–February 2015 period. Because residuals are heteroskedastic, inference uses HC3 robust standard errors, which correct the standard errors without changing coefficients, fitted values or forecasts.

READING THE FIGURES

Promotion uplift – the percentage sales effect of a regular daily promotion under the model's stated reference conditions, obtained by combining the relevant main and interaction coefficients before transformation.

Store baseline – the persistent trading level of a location, captured by store fixed effects.

Prediction interval – a planning range around the point forecast, treated as decision support rather than a guaranteed minimum or maximum.

Agnes & Co. RETAIL ANALYTICS

COMPANION TO ROSSMANN STORE SALES FORECAST · MMA 860 · AUGUST 2026